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RESEARCH-ARTICLE

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# GenShare: A Blockchain-Based Genomic Data Sharing Platform

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Every day, hundreds of gigabytes of data are produced due to the exponential growth of next-generation sequencing and omics technologies. By combining omics data with other data types, such as electronic health record data, panomics research is actively attempting to uncover novel and potentially useful biomarkers. For the effective analysis of high-throughput-derived omics data, it is imperative to establish robust and reliable platforms that prioritize ethical considerations while effectively managing privacy, ownership concerns, and the responsible sharing of data. The GenShare model was proposed to provide an efficient platform that fits these needs. GenShare is a hybrid platform that utilizes blockchain technology. Paillier's homomorphic encryption scheme in tandem with Intel Software Guard Extension (SGX) serves to enable the sharing of genomic data, execution of count queries, and statistical analysis of genomic data while preserving privacy and avoiding compromise of sensitive information. The objective of this paradigm is to confront security and privacy concerns through the integration of homomorphic encryption and SGX, addressing additional challenges associated with Hyperledger Fabric and Ethereum. In pursuit of this objective, the implementation of the system involved establishing the Hyperledger Fabric network, with various workloads employed to assess the network's efficiency. Consequently, it was hypothesized that the new GenShare model would enhance the data collection and dissemination cycle and serve as a proficient platform catering to the needs of its users.

CCS Concepts: • **Security and privacy** → **Management and querying of encrypted data**;

Additional Key Words and Phrases: Genomic Data Sharing, Blockchain, Homomorphic Encryption, Software Guard Extension (SGX), Ethereum, Hyperledger Fabric

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## 1 Introduction

A human has around 23,500 protein-coding genes. Genes hold the information that defines an organism's properties. A genome is the complete collection of genetic instructions for an organism, and genomics is the field that sequences and analyzes an organism's genome [1]. Initially, the sequencing procedure required a considerable amount of time and computation. However, the discovery of **Next-Generation Sequencing (NGS)** has changed genomics research [2]. The sequencing of the entire human genome now takes around 1 day, thanks to NGS. As a result, new genomic research is required to interpret the genome, and the amount of genomic

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data has increased dramatically. Because it describes the relationship between diseases and the causative gene or mutations, bioinformatics research has begun to play a vital part in human life. By evaluating omics or **Genome-Wide Association Studies (GWAS)** [3] data using AI-based algorithms, diseases are predicted and prevented before they become a serious threat. Based on genetic, environmental, and lifestyle factors, these therapies and preventions are being customized for each individual.

Especially in cancer, this method is preferred by doctors and researchers [4, 5]. To take advantage of all, genomic data should be shared easily. However, there are some problems in genomic data sharing [6, 7]. Privacy and security of data [8, 71], cost of analysis [9], data ownership and gathering [10], and handling massive amounts of data [11] are the primary issues in genomics data sharing. To address challenges related to data privacy and security, a variety of privacy-enhancing measures have been proposed, with notable examples including **Secure Multi-Party Computation (SMC)** **Homomorphic Encryption (HE)** [13]. However, other parts of the problem are still unavoidable for bioinformatics research, especially the data collection and distributed analysis issues. Genomic data should be easy to share and collect, and computations on them should be common. This would create a thriving ecosystem for all institutions, doctors, patients, and researchers involved in bioinformatics research. In the meantime, **Blockchain (BC)** technology has evolved to construct an immutable, trustworthy network without a central authority. It has recently attracted interest in a variety of fields [14], including genomics, due to the unique answers it offers to certain issues [15, 16].

In Table 1, we classify genomic uses of BC [17]. The goal of BC-based markets for genetic data is to eliminate the need for intermediaries and provide consumers authority over their data. Different incentive forms are offered to individuals who sell or rent their genetic data. Cryptocurrency is the most prevalent reward employed on these platforms. Genecoin [18] represented the first attempt to use BC in genomics when it was established in 2014, according to our research. The firm offers sequencing services via third-party laboratories. The resultant DNA sequence is subsequently encrypted and stored in the Bitcoin [18]. Numerous more marketplaces followed Genecoin to enable genomic data owners to trade their data for cryptocurrencies (tokens), which can then be traded on cryptocurrency exchanges. Other marketplaces differentiate themselves by providing additional services; for instance, Shivom [62] and Zenome [63] offer pipelines and computer resources for analyzing genetic data supplied to their platform. By paying costs in cryptocurrency/tokens, researchers can perform their studies using the supplied pipelines. In addition, data integrity and traceability are recognized as essential services provided by several genetic marketplaces. Genomes.io [64] enables customers to securely store and manage their DNA data from the moment it is sequenced until it is stored on the BC. Users are provided with a private access key that is stored only on their mobile devices and is used to govern data access. Genobank [65] also investigates the use of **Non-Fungible Tokens (NFTs)** for data portability and traceability. The suggested solution involves assigning a unique NFT to each human sample and broadcasting the information on a public BC. The NFT is applied for a variety of purposes, including determining which lab is capable of genome sequencing and monitoring data utilization. The selected study on non-commercial BC applications focuses mostly on the sharing of genetic data for the development of science. These research belong to one of the following categories of BC use cases: data exchange, data analysis, data storage, and access control.

While each publication is categorized based on its major research subject, overlaps are conceivable. For instance, one study focused on the sharing of genomic data to complete analytic tasks on that data, while another study focused on BC-based storage for the goal of sharing saved genomic data. The bulk of the recognized publications focused on utilizing BC to support/develop systems for multi-organizational or worldwide data exchange. In [66], multiple system architectures and platforms for exchanging genomic data employing various BC features and advantages were presented. Combining machine learning with BC was proposed in [67] to accomplish decentralized machine learning. Multiple businesses trained predictive models in [67], using BC for secure, transparent, and decentralized process coordination. The authors of [68] studied storage alternatives for Sequence Alignment Map files, which may reach tens of gigabytes in size. This was done by utilizing data compression

Table 1. Classification of Genomic Uses of BC Technology

| Main Category                      | Focus                                                                                                     | References |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|------------|
| Commercial                         | Encrypted DNA storage/backup in the Bitcoin BC                                                            | [18]       |
|                                    | Secure storage, selling access to genetic data, and buying genetic services                               | [63]       |
|                                    | Secure personal DNA vault and financial benefits from (anonymous) contributions to medical research       | [64]       |
|                                    | Control over genomic data with DNA crypto wallet, and secure platform to process the data                 | [65]       |
|                                    | Pay-per-use marketplace for genomic data, bioinformatics platform with analysis pipelines for researchers | [62]       |
| Data Sharing                       | Secure genomic data-sharing platform                                                                      | [66]       |
| Data Analysis                      | Decentralized privacy-preserving predictive model learning                                                | [67]       |
| Storage                            | Storing raw genomic data on-chain                                                                         | [68]       |
| Data Management and Access Control | Access and consent management of personal genomic data                                                    | [69]       |

techniques and a private BC network to create a unique data format. Another group of publications [69, 70] focuses on the use of BC as a way to grant/revoke access to genetic data housed elsewhere in the form of consent management. If we summarize the studies, each has advantages and disadvantages compared to the others. These deficiencies can be summed up as follows: Firstly, neither the means for verifying the data's quality and integrity nor the resolution of the issue of mistrust of authorities are discussed. Scalability is an issue for which there is no specific solution on the platform. Genomic data tend to be very massive and difficult to replicate on each node. Thus, this negatively impacts the speed and scalability of a fully distributed system. Alternately, other researchers propose the use of BC to manage health data that is kept off-chain as opposed to storing actual patient data, while a derived BC stores lightweight metadata about the off-chain files. Generally, cloud or IPFS is used for off-chain storage. While cloud-based systems break the decentralized structure of the system, IPFS increases the computational cost. In research employing permissionless BC, it is unclear how users might access or participate in the network. Also, even if data is encrypted and exchanged inside the system, it has not been considered that a malicious owner may send worthless data to a recipient.

In research utilizing permissioned BC, neither who can join the system nor how the system's permissions are determined are described in depth. In addition, system administrators are presumed to be completely dependable, and the impacts of hostile agents on the system have not been investigated. It does not define how data is transferred during data sharing, even though many systems allow the recipient to review the data's characteristics before data sharing. According to our findings, platforms have moved in the direction of resolving some of the lingering issues around genomic data sharing that they deem crucial. However, to ensure full efficiency, the whole genomic data sharing problems should be considered in detail. In this direction, we aimed to develop a new platform that eliminates the disadvantages of existing methods and produces solutions for privacy and security of data, cost of analysis, data ownership and gathering, and handling massive data size. This article proposes GenShare, which is an innovative dual BC-based platform for managing both data sharing and computation. GenShare protects the privacy and security of genomic data by combining HE [19–21] with **Software Guard Extension (SGX)** and BC. It also offers BC-based solutions for the cost of analysis, data ownership, how to get the data, and a wide range of other data-related problems [22–24]. The hybrid GenShare platform comprises **Secure Compute Nodes (SCN)**, data owners, and researchers for rapidly transferring genomic data and performing computations on it, utilizing the different viewpoint approach offered by Ethereum [25] and Hyperledger Fabric [26]. In GenShare, data owners will privately share their indexes on the BC [27], whereas researchers are a public



community that wishes to access shared data. Thus, GenShare has selected both [28] Hyperledger Fabric and Ethereum to fulfill these needs.

GenShare came about within the scope of sharing genomic data, but the components it builds on, such as HE, **Trusted Execution Environments (TEEs)**, and distributed ledgers have even wider scope synergies across various fields. HE has allowed credit scoring and fraud detection to take place without underpinning data revealing in finance [72, 73]. In an even broader scope, SGX is used in cloud computing and secured data analytics platforms such as Ekiden and Oasis Labs that provide off-chain computations which are trusted [74]. IPFS and Filecoin are examples of distributed systems which provide data storage frameworks that are decentralized, immune to data corruption, while solving the data availability problem and preserving the integrity of the data [75, 76]. These use cases illustrate that the balance between privacy and usability is not exclusive to genomics. GenShare has tried to use techniques from many fields to showcase an architecture designed for genomics but usable in other domains sensitive to privacy concerns. The operation of the study is summarized in the flowchart in Table 2. The main contributions of our study can be summarized as follows:

- (1) Current proposed systems have not given sufficient consideration to obtaining genomic data and data verification procedures [29]. GenShare focuses on this crucial issue and gives use-case scenarios regarding it.
- (2) When developing the GenShare model, system design is separated into two categories: (i) incorporating nodes within the GenShare system and elucidating the connections between these nodes, and (ii) acquiring genomic data. The system structure's classification aimed to investigate relevant solutions for each step and identify the most suitable techniques for GenShare's system requirements. We believe that the different solutions given in this methodology will shed light on several additional systems.
- (3) In the field of bioinformatics, to the best of our knowledge, the combination of different BC technologies and the provision of interactions between them have not yet been investigated and illustrated. In this work, the architecture of a hybrid BC system that works well is explained in detail.
- (4) Due to HE and SGX [30] technology, genomic data that has been decrypted cannot be accessed by anybody other than the data owners. GenShare's strong SGX-enabled servers make it simple to do statistical analysis and count query operations on genomics data. These servers will participate in both the Ethereum network and the Hyperledger Fabric network.
- (5) The data owners can assume control and manage permissions for their data while sharing it with third parties. The GenShare platform also enables data owners to get benefits, such as money, for sharing their data, and to encourage people to contribute to research. We believe that all of these positive changes will motivate data owners to volunteer their information so that sufficient data may be gathered for scientific research.
- (6) For performance analysis of the system, the Hyperledger Fabric network is constructed. The system's performance has been demonstrated at a variety of workload rates. Typically, implementations and tests of studies are conducted within the local network. In contrast, in GenShare tests, nodes communicate via the global network using API calls to determine the cost of communication. Platforms similar to GenShare cannot operate on local devices, and delays on the global network will undoubtedly impact the system.

The rest of the article is organized as follows: The Methods section describes the fundamental techniques used to construct GenShare. In GenShare Model section, we formulate the processes and describe the proposed platform in depth. The Implementation part describes the implementation environment, whereas the Security Analysis section discusses possible security attacks and solutions. Then, we show and discuss the performance analysis of the GenShare platform in the Results and Discussion section. The Conclusion section summarizes the proposed model's key contributions and future directions.

Table 2. The Operation of the Study

|                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Step 1.</i> Investigation of the importance of genomic data sharing                                                                                                       | Bioinformatics research has become important because genomics links illnesses to their genes or mutations. AI-based systems predict and prevent illnesses using -omics or GWAS data. Doctors and researchers recommend this technique, especially for cancer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <i>Step 2.</i> Identifying problems that prevent genomic data sharing                                                                                                        | Genomic data should be easily disseminated to benefit everybody. However, genomic data sharing has some issues. They are mainly privacy, security, cost of analysis, ownership, data harvesting, and management of enormous volumes of data. SMC and HE are two used privacy-enhancing technologies for providing data privacy and security. However, bioinformatics research must still deal with data collecting and dispersed analysis challenges. BC technology now creates an immutable, trustworthy network without a central authority. Due to its unique solutions, genomics and other sciences are interested in it.                                                                                                                                                                                                                                                                                    |
| <i>Step 3.</i> Examining BC-based genomic data sharing systems                                                                                                               | We categorized the current studies into non-commercial applications and BC-based commercial genomic marketplaces. Non-commercial applications employ BC for data exchange, analysis, storage, and access control. Each study has pros and cons. At the end of the review, these deficiencies were determined in order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <i>Step 4.</i> In line with these implications, determining system requirements and components that will play a role in achieving the BC-based genomic data sharing platform | We wanted to create a platform that solves data privacy and security, cost of analysis, data ownership, data collection, and large data management. This article proposes the GenShare which is an innovative dual BC-based platform for managing both data sharing and computation. GenShare protects the privacy and security of genomic data by combining HE with SGX and BC. It offers BC-based solutions for analysis cost, data ownership, data access, and other data issues. Researchers, data owners, and secure computing nodes use Ethereum and Hyperledger Fabric's diversified perspective to effectively exchange genetic data and compute it on the hybrid GenShare platform. In GenShare, data owners will privately share their BC indexes, while researchers are a public community that wants to access shared data. GenShare chose Hyperledger Fabric and Ethereum to meet these objectives. |
| <i>Step 5.</i> Achieving the GenShare                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Step 6.</i> Explanation of the methods used in the platform                                                                                                               | The technologies used in the platform, how they work with the system, and how the platform is put together as a whole are all explained. These methods are Hyperledger Fabric, Ethereum, HE, and SGX.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <i>Step 7.</i> Security Analysis of the GenShare                                                                                                                             | Security problems that may occur on the platform and their solutions are explained step by step.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>Step 8.</i> Implementation of the GenShare                                                                                                                                | The Hyperledger Composer Business Network is implemented and tested for performance analysis. The name of this business network was determined as "agunetwork." The outcomes are displayed with varying numbers of nodes and transactions. High-performance hardware resources are required to make the whole system work. In this study, the first part of the application was carried out due to the insufficient capacity of the hardware resources we have.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Step 9.</i> Demonstrate and discuss system performance analysis                                                                                                           | AvgTL and TT metrics were used for the performance measurements. The "agunetwork" was measured twice, while the system contained 100 and 500 nodes. Different workload rates are determined for system performance evaluations. They are: <ul style="list-style-type: none"> <li>(i) AvgTL and TT results for the participant and asset creation.</li> <li>(ii) AvgTL and TT results for the requested data transaction.</li> <li>(iii) AvgTL and TT results for the requested data transaction, by collisions.</li> <li>(iv) AvgTL and TT results for the requested data transaction, with 100% collisions.</li> </ul>                                                                                                                                                                                                                                                                                          |

AvgTL, Average Transaction Latency; TT, Transaction Throughput.

## 2 Methods

### 2.1 BC Technologies

The BC is an immutable transaction ledger that provides a secure, decentralized system without central authority [31, 32]. Each transaction in the system is represented by cryptographically signed blocks [33], and network users are responsible for transaction verification [34].

Choosing the most appropriate BC platform is a crucial step in planning and developing a new BC-based project. Public and private ledgers can be created using BC [35]. Hyperledger Fabric and Ethereum are notable instances of private and public BCs, respectively. Ethereum and Hyperledger Fabric were created for very different reasons [36]. Since Ethereum is designed for a public BC, it is supposed to be entirely transparent, permissionless, and accessible to everyone. In contrast, Hyperledger Fabric offers solutions for permissioned BCs that enable security and anonymity. Its ledger is only available to authorized individuals. Therefore, it is possible to construct many channels and connect only a subset of users to them; the data cannot be accessed by users who are not enrolled as a result, and confidential information may be shared without broadcasting the transaction to the rest of the network. The network configuration pre-defines the responsibilities of nodes in the Hyperledger Fabric consensus process.

A smart contract is a set of rules that are executed on the BC. As the software representative of its users, it automatically carries out specific responsibilities and tasks when the conditions are met. The objective of smart contracts is to preserve data, manage contracts and relationships, and provide functionalities and complex authentication for other contracts [37, 38]. Nearly all BC-based healthcare and genomics applications utilize smart contracts [39, 40]. GenShare advocates the incorporation of smart contracts within the proposed framework, citing their diverse benefits, including the automated execution of obligations and the uncomplicated management of data access permissions and associations. Hyperledger Fabric and Ethereum are the most prominent smart contract-based BC platforms. However, Hyperledger Fabric uses chain codes. While a smart contract delineates the executable logic responsible for generating novel information and disseminating it across the distributed ledger, administrators frequently employ chain code to categorize analogous smart contracts for deployment and to architect the foundational framework of Fabric at a low level.

In GenShare, data owners will privately distribute the indexes of their data on the BC [27], whereas researchers are a public community that wishes to access shared data; thus, two chains are preferred. GenShare has selected both [28] Hyperledger Fabric and Ethereum to fulfill these needs. In the GenShare framework, data owners and hospitals are integrated within a closed system of known participants. They function as private users of the system and employ Hyperledger Fabric. Wrapped consensus algorithms are known as the Raft algorithm [77]. In this case, it provides crash fault tolerance, low latency, and high throughput, which are necessary to satisfy the performance requirements of such a permissioned system. GenShare adopts an Ethereum BC for its open portion facing computation demand from researchers, which is set to have a Proof-of-Authority consensus mechanism [78]. With this design, only previously allowed nodes will be able to participate in the validation of transactions which serves both performance and security purposes. To mitigate the potential communication and synchronization overhead that may arise due to the execution of two distinct BCs, GenShare implements an asynchronous interaction model. Hyperledger Fabric and Ethereum run independently per their assigned tasks and responsibilities, while the SCNs serve as intermediaries that manage the queuing of computation requests and the swapping of data indices between the two chains. The structure is less demanding in terms of needing to conduct real-time synchronizations and more accommodating towards modular scalable design. The management of two distinct BC infrastructures comes with increased system complexity and additional computational tasks, particularly around identity management, smart contract incorporation, and dataflow synchronization. All these problems become even more difficult in scenarios when SCNs have to interface with both BC at the same time. GenShare solves these problems with its modularity which eases the burden while keeping the system efficient. The selection of these two BC technologies is further informed by the additional factor that Hyperledger Fabric

facilitates Ethereum smart contracts through the incorporation of the Ethereum Virtual Machine chain-code plugin. It means that the proposed solution can interact between Hyperledger Fabric and Ethereum. This is significant since each BC project targets a specific sector and, as in the case of the GenShare project, may need the use of many technologies in combination, depending on the desired qualities. In the realm of cross-chain interaction spanning public, private, and consortium BCs, as well as BC interoperability [41], there is ongoing scrutiny of the amalgamation of mechanism-related considerations.

We may primarily classify the difficulties associated with sharing genomics data as follows: (1) data collecting, (2) ownership, (3) data sharing, and (4) analysis cost. Rather than establishing a direct connection between the data owner and the researcher, conventional business models for genomic data sharing incorporate the involvement of intermediary organizations. As a result, the cost of analysis rises, the data owner loses control over data access restrictions, and it becomes impossible to profit from data sharing. Due to the advantages of BC [22–24]: (1) middleman can be eliminated, and data owners and researchers can communicate directly and earn some money, (2) data owners can manage access permissions to their data and use smart contracts to authorize data access, (3) data owners can anonymously share their data, and (4) with the high computational power of BC, analysis of a huge amount of genomic data that takes trillions of CPU hours can speed up. The GenShare platform based on a dual BC structure shows how to solve all these problems as stated.

## 2.2 Partially HE (PHE)

SMC and HE are the most prevalent privacy-enhancing methods used to overcome genomic data privacy-security challenges. SMC facilitates concurrent computation involving two or more parties, allowing them to obtain the results without any party gaining access to the inputs of the others. Despite its numerous advantages and applications [43, 44], SMC is currently impractical for most applications that require (near) real-time performance. The HE allows addition and multiplication operations to be conducted on ciphertexts and generates encrypted outputs that correspond to the plaintext operations [45]. There are three types of HE methods [46, 47] based on the number of operations that can be done on the ciphertexts:

- PHE: PHE allows only one type of operation (addition or multiplication), an unlimited number of times.
- **Somewhat HE (SWHE)**: SWHE permits certain categories of operations, albeit within a constrained frequency.
- **Fully HE (FHE)**: FHE allows an unlimited number of operations, with an unlimited number of times.

Various genomic computations are carried out today using HE [48, 49]. The PHE method is sufficient for statistical analysis of genetic data's frequency count [50]. SWHE should be preferred in complex operations such as pattern matching and searching [51]. The size of ciphertexts increases with each HE operation [52] as the number of operations that a scheme may execute increases. Hence, the conceptual simplicity of constructing an FHE scheme contrasts with the inherent challenges involved in its practical implementation. GenShare has opted to employ Paillier's encryption system, which is PHE and is summarized in Figure 1.

This method is used to solve the data privacy and security problem in combination with BC and to use the additive homomorphic property to do safe computations on genomic data.

## 2.3 Intel SGXs

SGX constitutes an architectural security feature integrated into Intel processors, allowing the establishment of secluded memory regions referred to as enclaves. Through the utilization of SGX, the confidential codes and data within these enclaves are isolated and safeguarded from privileged modules. The hardware-supported access control restricts entry to the enclaves. Therefore, the SGX may perform secure computations on confidential data provided by untrusted parties [53]. Combining the SGX and HE is another method of computing [54]. Thus, although the SGX protects data, HE ensures data privacy. The summary of this situation is shown in Figure 2.

| Key Generation                                                                                                                                                                                                                                                                                                                                                          | Encryption of m                                                                                                                                                                                                           | Decryption of c                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. <math>n=pq</math>, the RSA modulus</li> <li>2. <math>\lambda=\text{lcm}(p-1, q-1)</math></li> <li>3. <math>g \in \mathbb{Z}/n^2\mathbb{Z}</math> s. t. <math>n \nmid \text{ord}_{n^2}(g)</math></li> <li>4. Public-key : <math>(n, g)</math>, secret key: <math>\lambda, \mu</math></li> </ol>                                | <ol style="list-style-type: none"> <li>1. <math>m \in \{0, 1 \dots n-1\}</math>, a message</li> <li>2. <math>h \in \mathbb{Z}/n\mathbb{Z}</math></li> <li>3. <math>c = g^m h^n \bmod n^2</math>, a cipher text</li> </ol> | <ol style="list-style-type: none"> <li>1. <math>m = L(c^\lambda \bmod n^2) L(g^\lambda \bmod n^2)^{-1} \bmod n</math></li> <li>2. The constant parameter, <math>L(g^\lambda \bmod n^2)^{-1} \bmod n</math> or <math>L(g^\mu \bmod n^2)^{-1} \bmod n</math> where <math>g = 1+n \bmod n^2</math> can also be recomputed once for all.</li> </ol> |
| <p><b>Example:</b> Suppose there are two ciphers. If <math>\text{CipherText1} = g^{m1} X_1^n \bmod n^2</math> and <math>\text{CipherText2} = g^{m2} X_2^n \bmod n^2</math>, <math>\rightarrow \text{CipherText1} \cdot \text{CipherText2} = g^{m1} X_1^n \cdot g^{m2} X_2^n \bmod n^2</math> and additive property is: <math>g^{m1+m2} (X_1 X_2)^n \bmod n^2</math></p> |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                 |

Fig. 1. The Paillier algorithm.

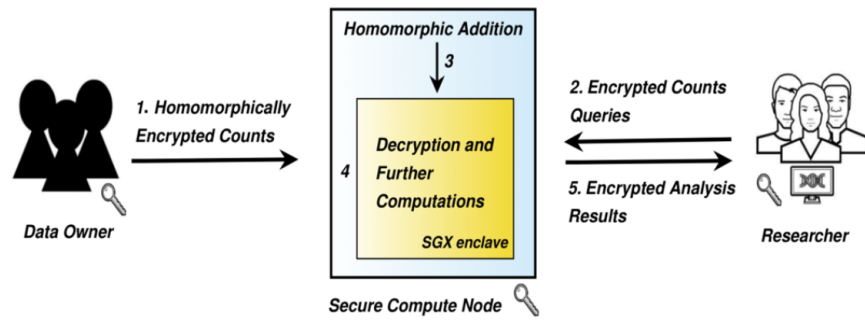


Fig. 2. Combining SGX with additively HE.

By employing a fusion of Paillier’s methodology and SGX technology, various computations utilizing GWAS and NGS data can be executed. These include assessments such as Disease Susceptibility, the generation of a Contingency Table, and the Statistical Analysis of Genomic Data [55] as Linkage Disequilibrium and Hardy-Weinberg Equilibrium [19–21].

In the GenShare platform (as described in Section 3), genomic information like SNPs are represented as integer vectors that undergo HE by means of the Paillier scheme. There is a hashed identifier for each data vector that can be used as an index for secure retrieval, and these identifiers are stored as indexes. These indexes are given to the SCNs that perform computations like counting the frequency of alleles, generating contingency tables, and running statistical tests like Hardy-Weinberg Equilibrium and Linkage Disequilibrium. Although this design allows for privacy-preserving statistical analysis, it does not capture more complex pattern recognition or multi-dimensional filtering queries. We would like to emphasize these practical lacunae, as they are beneficial for future research avenues. Due to the previously described advantages, in this study, combining SGX with Paillier’s HE has been chosen to execute calculations on GWAS and NGS data.

### 3 GenShare Model

GenShare is an innovative BC, HE, and SGX-powered platform for the exchange and analysis of genetic data. The system architecture consists of two distinct classes: (i) the incorporation of nodes within GenShare along with their relationships, and (ii) the acquisition of genomic data.

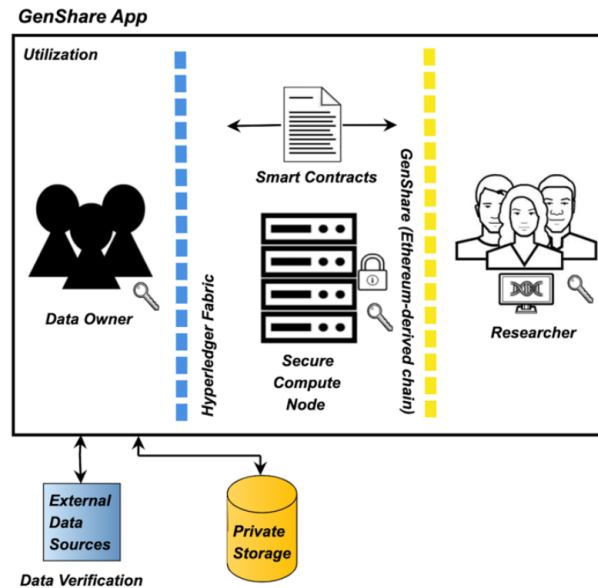


Fig. 3. Overview of the GenShare model.

### 3.1 The Incorporation of Nodes within GenShare along with Their Relationships

GenShare consists of three separate roles, as seen in Figure 3, of nodes: (i) data owners, (ii) researchers, and (iii) SCNs.

- Data Owners: Individuals seeking to share their genetic data can delineate and configure the access and privacy permissions associated with the data. Sharing their data generates revenue for data owners. Using the Hyperledger Fabric network and the GenShare application, they may secretly swap their data index.
- Researchers: They are mostly university members. Researchers acquire genetic data from data owners through SCNs, subsequently analyzing the data without directly accessing its content on the SCNs. They are active on the GenShare BC, a network derived from Ethereum.
- SCNs: These robust servers endorse SGX technologies and perform computations on homomorphically encrypted data. They stand as the foundational elements of the system and are supervised by designated academic institutions. By utilizing both the Hyperledger Fabric and GenShare networks, they build a connection between the data owners and the researchers without being able to view the contents of the data.

In GenShare, data owners will discreetly share their data indexes on the BC [27], whereas researchers are a public community that wishes to access shared data. Thus, GenShare utilizes two separate BC systems, (i) Ethereum and (ii) Hyperledger Fabric. The GenShare chain contains researchers, whereas the Hyperledger Fabric network has data owners. Each BC comprises protected computational nodes facilitating intercommunication amidst the system. Data types and computation costs may be statically defined by GenShare or dynamically selected by system users. Users are allowed to make their own decisions under dynamic regulation. But to make sure that everyone in the GenShare network gets the same amount of work, fixed prices are set for each type of data and calculation. Wherever they choose, data owners in GenShare can store their data in homomorphically encrypted form. The properties of their data are first defined when they enter the network. Instead of exchanging actual data, they share data indexes with SCNs during the process of data sharing. By



transmitting indexes, SCNs obtain homomorphically encrypted data. The hospital is obligated to corroborate the ailment specified by the data owner within the data attributes, while the SCNs must validate the hash records about the lab's data. This verification procedure was described in depth in the section titled "Acquisition of Genomic Data." If the data is validated, the funds will be paid to the data owner's account and the researcher's calculations will be carried out. If the data cannot be authenticated, the owner of the data will be banned and punished.

The declarative pay-per-use model within GenShare's architecture pays the user for the precise amount of data they use. The GenShare system does not operate like traditional centralized data marketplaces where the proprietary algorithms stored in a black box are executed. Rather, GenShare employs Ethereum smart contracts to automate and guarantee payment to data owners after their data has been validated and computations done. Eliminating trusted intermediaries enhances transparency, reduces costs, and operational overhead. This model, by providing every genomic data sharing platform incentive architecture, addresses one of the most significant gaps in platform design: primary motivator for voluntary data contribution is missing. Moreover, embedding this model in a public BC enables the creation of auditable records of data usage and transactions—essential for sustainable trust among stakeholders.

The adoption of both Hyperledger Fabric and Ethereum is not arbitrary and each technology choice is tailored to the context of GenShare's dual goals of privacy-preserving data management and transparent, researcher-facing computations. Hyperledger Fabric was selected because of its permissioned structure, which allows for refined access control and validates trust within subsets of hospital data providers. On the other hand, Ethereum allows researchers to work in a transparent and decentralized environment where computation requests, smart contracts, and results can be verified and publicly accessed. A single-chain solution was found lacking either in privacy, as in Ethereum's case, or transparency and scalability in open users as was Fabric's case. GenShare's approach to modularity and scalability is achieved through asynchronous inter-BC unlinking using SCNs which minimizes cross-BC synchronization overhead. This architecture guarantees the simultaneous achievement of privacy, security, performance, and data openness, which would be impossible to realize with a single BC configuration.

The participation of researchers in GenShare is contingent on two requirements. These conditions define whether the computations that researchers want to make on data will be made public and whether a researcher will execute actions on the system individually or not. The BC, which involves researchers, can function as an off-chain [56] to ensure confidentiality. At the same time, if at least three researchers assemble, they might request that the data be processed with threshold cryptography [57]. Therefore, unnecessary calculation requests are eliminated, making the system safer. After joining the system, the researchers construct a smart contract on the Ethereum-derived chain with the appropriate data properties and computation type. Upon examination of the researchers' agreement, the SCN scrutinizes the Hyperledger Fabric to identify suitable data owners. Upon identification of the relevant data owners, the associated fee is expeditiously gathered from the researchers. Furthermore, a selection protocol may be devised by system requisites to choose a node possessing the requisite property data from a pool of potential data owners. Upon verification of the data owner by the secure computing node, computations are executed, and the resultant output is transmitted back to the researcher. Certain universities serve as SCNs in the system. To integrate GenShare, they must endorse SGX technology, given that the shared data is encrypted using HE. Their responsibilities encompass data verification, execution of computations as requested by researchers, and the subsequent submission of results. Secure computing nodes interpret researchers' requests and retrieve the relevant data accordingly. After discovering pertinent data, they are compensated by researchers. Suppose an issue arises in the SCN before the result reaches the researcher. Under such circumstances, the transaction might be allocated to a new computation node, or the reversal mechanism could refund the researcher's funds. Similar to any other system, GenShare is susceptible to trust-related concerns. To mitigate this risk, the system may incorporate nodes tasked with validating computation results.



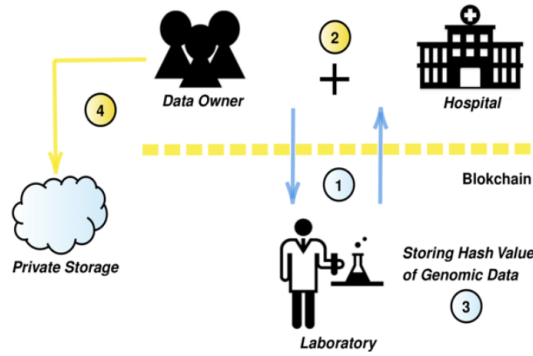


Fig. 4. The process of obtaining genomic data.

### 3.2 The Acquisition of Genomic Data

Currently, relevant laboratories are sequencing the genome. In the field of disease research, genome sequencing is often undertaken by laboratories, but individuals occasionally desire to discover the maps of their genes. Individual samples should be sent to laboratories to obtain relevant results. This procedure involves either mailing the samples to the laboratory or having the individuals travel straight to the labs. At this stage, a person's privacy cannot be guaranteed because the laboratories are aware of their identities. Briefly, complete anonymity cannot be discussed until people have portable technologies to sequence their genomes outside of a laboratory. In the final phase of receiving genomic data, laboratories conduct studies on samples using the sequencing process and convey the results to individuals. Malicious users can be found in any system or organization, even though labs usually delete the data after the results are shared with the owner.

At the time of sharing an individual's genomic data, three distinct challenges must be addressed: (i) disease verification, (ii) data security, and (iii) disease confirmation. Hospitals mostly sequence genetic data for illness studies. Before sharing their data, the data proprietor provides the researcher with data attributes, requiring authorization from the hospital to validate the existence of a disease within the data characteristics. The implementation of a data authentication system, capable of discerning the authenticity of user-provided data, becomes imperative. It is postulated that the engagements between the lab and the data proprietor are recorded on the BC, exemplified in Figure 4, segment 1. In this instance, (i) the lack of confidence in laboratories is erased, (ii) the laboratory from which data is collected seems transparent, and (iii) the laboratory is banned from selling data to third parties. Simultaneously, by including hospitals in the BC, the disease verification mechanism can be provided directly.

## 4 Security Analysis

The construction of the proposed model involves categorizing the system structure into distinct segments, wherein various alternative solutions for each category are considered. The optimal option for each category is then chosen based on the specific requirements of the GenShare system. This categorization is implemented to mitigate potential security issues. Within the GenShare framework, genetic data is stored securely by the owner in a private database, utilizing HE. Subsequently, computations on encrypted data are conducted securely through the application of SGX technology. That ensures the privacy and protection of data. The Hyperledger Fabric network provides an index of encrypted data. GenShare initially blocks access to the data index for all users excluding the data owner. Thus, when SCN pays the fee for data to the data owner, it instantly gains the ability to view the data index. This prohibits access permission from being granted to unauthorized individuals [60]. As a data verification technique, GenShare advises that laboratories save the data hash values. By comparing the hash

values of the APIs, the SCN can simply determine if the data is authentic or not, preventing data manipulation. In the event of a user engaging in malicious activities, the system can suspend their account, and comprehensive logging of all attacks is maintained as evidentiary documentation. To preempt denial-of-service attacks [61], a preventative measure is implemented wherein, upon the collaboration of at least three researchers, a collective request to process the data is facilitated through the utilization of threshold cryptography. Certain colleges in the system provide secure computing nodes, although there may be trust issues in each network. To prevent this, the system might contain nodes responsible for checking the SCN's computation results.

Illustrated in Figure 4, segment 2, the establishment of partially shared keys enables data owners and hospitals to operate as a unified node, necessitating mutual consent before engaging in data exchange. To prevent security breaches, the approval method may need at least three physicians from a hospital to acknowledge the illness. As a data verification strategy, GenShare recommends laboratories save the hash values of data, as depicted in Figure 4, part 3. By comparing hash values, secure computing nodes may simply determine whether the data is authentic when necessary. Even if the identity of the owner of genomic data is unknown, it is feasible to gather information about her/his identity and ancestry by analyzing the owner's data. Therefore, guaranteeing the security of genetic data is of the utmost importance. As illustrated in Figure 4, part 4, the most efficient solution for this is for data owners to keep their data anywhere in homomorphically encrypted form and share the sole data index. Therefore, calculations are performed without knowledge of the data's content, and only the data's owners may view the data's content.

Although Intel SGX offers a hardware-based TEE which greatly enhances the privacy of data during its computation, it does have some shortcomings. Various research works have pointed out the possible side-channel attack vulnerabilities of SGX, including cache timing and speculative execution [79]. GenShare does face these risks, but the platform alleviates them through periodic remote attestation and enclave code audits, non-branching control flow homomorphic calculation methods, and the restriction of operations on the enclave to low-branch-count routines. We accept these risks, especially the ones pertaining to ARM Trust Zones and deem as an enhancement the possible future integration of other TEEs, such as AMD SEV, or the application of techniques involving computation verification [80].

## 5 Implementation

Implementation of the GenShare system can be examined in three parts, as follows:

- (1) Enabling communication between data owners and secure computing nodes.
- (2) Executing secure computations on shared data.
- (3) Facilitating communication between secure computing nodes and researchers.

In this study, the Hyperledger Composer Business Network, also known as “agunetwork,” is implemented and tested. The outcomes are displayed with varying numbers of nodes and transactions. High performance hardware resources are required to make the whole system work. In this study, the first part of the application was carried out due to the insufficient capacity of the hardware resources we have. Hyperledger Composer serves as a development platform designed to streamline the creation of applications for the Hyperledger Fabric BC. Agunetwork has been established on two distinct physical nodes that are linked to the global network. Node 1 is powered by a server with four vCPUs, 8 GB of memory, and 50 GB of storage capacity. Node 2 is powered by a server with four vCPUs, 8 GB of RAM, and 50 GB of storage capacity. Ubuntu 18.04 stands out as the optimal operating system due to its compatibility with Hyperledger Fabric 1.4. The deliberate choice of a global network for communication among physical nodes is driven by the aim to yield authentic results in the evaluation of agunetwork performance. Participants in the agunetwork are data owners and SCNs. Assets in the agunetwork are genomic data and data index. Transactions in the agunetwork are SCN creation, owner creation, assets creation, and data request. SCNs join the agunetwork with their affiliation information. DataRequest is a

transaction requested by SCN for the URL of data. If the prerequisites are satisfied, access is allowed to the data URL for the SCN.

There are three ways to offer data request transactions: smart contract, chain code, and modifying access rights. Although smart contracts provide a solution for users, the implementation of chain codes offers administrators a solution. The purpose of the two solution techniques is to automatically update the linked asset or participant information when the requisite circumstances are satisfied. However, the employment of these two approaches in GenShare generates more labor. As per the system specifications, each data owner possesses both genomic data and a URL asset. If GenShare had employed either of these two strategies, the owner of genomic data and URL assets would have undergone modification in the data request transaction to become a SCN. In such a scenario, the data owner would be precluded from reselling identical data; consequently, to facilitate its sale, they would be required to generate alternative assets for the same data. This imposes an additional burden on the agunetwork whenever data is shared. In addressing this obstacle, GenShare, rather than modifying the ownership of genomic data and URL assets, initially implemented limitations on access to URL assets. This measure authorized exclusive access to the data owner, effectively alleviating the pertinent issues. Before the Data Request transaction, SCN searches the agunetwork for the genomic data. When SCN discovers acceptable genomic data, it pays the data's cost. If the quantity of money supplied equals the cost of the genetic data, SCN may automatically read the related URL asset; otherwise, access remains restricted. Using all of these techniques, the agunetwork's additional workload was avoided.

To test API calls, Apache JMeter [58] was used with the Hyperledger Composer REST server. The Hyperledger Composer Rest Server is utilized to produce a REST API that can be simply consumed by HTTP or REST clients on a Hyperledger Fabric business network. Apache JMeter is a load testing and performance measurement tool for web applications and other services. Instead of a local network, API calls were transmitted externally over the global network. The reason why queries are sent across a worldwide network is because the actual GenShare program cannot function locally. Likewise, it sought to demonstrate the consequences of global network delays on the system.

## 6 Results and Discussion

This section will illustrate Hyperledger Fabric network performance for the provided transactions and workloads. Key performance metrics of the Hyperledger BC are Read Latency, Read Throughput, Transaction Latency, and **Transaction Throughput (TT)**. Since the operations in this study were transaction-based rather than read, **Average Transaction Latency (AvgTL)** and TT metrics were used for the performance measurements. Before displaying workloads, results, and test conclusions, these performance metrics will be clarified [59]:

- AvgTL: It denotes the duration between the submission of a transaction and the moment when the result becomes broadly accessible on the Internet. This encompasses both the propagation duration and any necessary settling time associated with the consensus method employed. In the context of this study, emphasis is placed on the AvgTL as the preferred metric.

AvgTL = subtracting the submit time from the total confirmation time (calculated individually for each transaction) and then dividing this value by the total number of transactions.

- TT: It is defined as the pace at which valid transactions are successfully committed by the tested BC system within a designated time frame. This rate is quantified on a network scale, typically measured in transactions per second.

TT = the total number of committed transactions divided by the total time duration in seconds.

The evaluation of the agunetwork occurred on two occasions, corresponding to system configurations of 100 and 500 nodes. In both scenarios, whether the network comprised 100 or 500 nodes, 10% of the nodes were designated as SCNs, while the remaining 90% functioned as data owners. Each system owner held both genetic

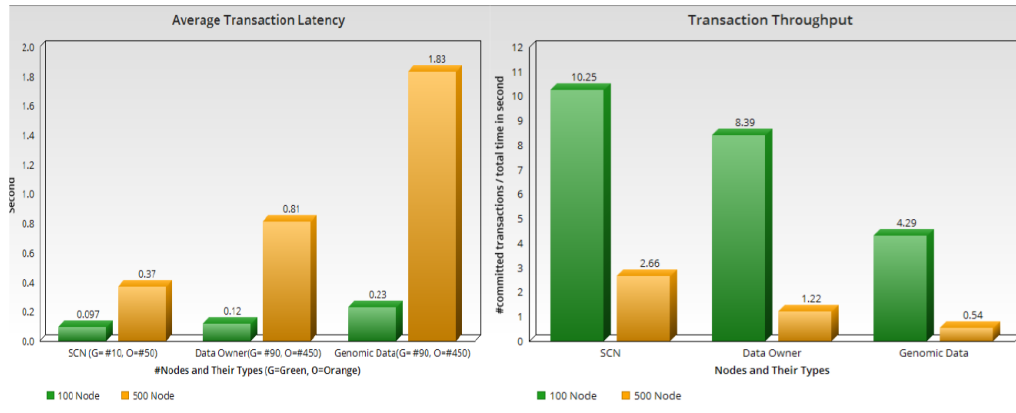


Fig. 5. AvgTL and TT results for the participant and asset creation.

data and a URL asset. It is imperative to emphasize, during the data analysis, that API queries directed to GenShare traverse the global network, not the local network. Consequently, the Internet speed exerted some impact on the results, hindering the discernment of a consistent growth rate among them. Before disclosing the results, many conditions must be specified. For system performance evaluations, several workload rates are calculated. The rates in Figures 5–8 were set by the constraints imposed by the agunetwork and the hardware on which it operates. The system supported a maximum of 900 concurrent transactions when operating on the agunetwork. These speeds may be modified by increasing the nodes' hardware capacity and by adjusting the Hyperledger installation parameters. Using agunetwork's hardware resources, the given ratios are the ideal benchmarks for assessing network performance. Initially, the examination focuses on the establishment of data owners, SCN, and assets for each workload. Figure 5 illustrates the AvgTL and TT associated with both participant and asset creation processes.

The introduction of the workload resulted in a fivefold escalation in the volume of transactions. As anticipated, with the increasing number of transactions, there was a corresponding rise in AvgTL and a reduction in TT. It is noteworthy to emphasize that the quantity of data owners and asset creations is equivalent. However, the temporal lags in transactions linked to the generation of assets are roughly twofold compared to those associated with the creation of data owners. The TT for asset creations is roughly half that of data owner creations, attributable to variations in the size of the transmitted messages. Specifically, 115 bytes, 294 bytes, and 489 bytes are transmitted during the creation of a SCN, data owner, and assets, respectively. Consequently, a rise in message size, including a rise in transaction volume, affects AvgTL and TT results.

Figure 6 presents the AvgTL and TT results for data request transactions based on a particular workload. In the process of node creation, the distribution of nodes in the agunetwork is 10% SCN and 90% data owner. On 100 nodes and 500 nodes, the following workloads are applied:

- (1) At the same time, 10% of the SCNs in the network purchase genomic data of 1, 5% and 10% of the data owners, respectively.
- (2) At the same time, 20% of the SCNs in the network purchase genomic data of 1, 5% and 10% of the data owners, respectively.

The outcomes observed for the configuration involving 100 nodes demonstrate that with an increment in the volume of transactions, there is a reduction in AvgTL coupled with an increase in TT. In a comparative context, an analysis of data from 500 nodes validates that, correspondingly, as the number of transactions expands, AvgTL experiences an increase, while TT undergoes a decline.

| For 100<br>Node | 1 SCN                             |                           | 2 SCN                             |                           |
|-----------------|-----------------------------------|---------------------------|-----------------------------------|---------------------------|
|                 | Average<br>Transaction<br>Latency | Transaction<br>Throughput | Average<br>Transaction<br>Latency | Transaction<br>Throughput |
| 1<br>Owner      | 0.15                              | 6.62                      | 0.14                              | 6.83                      |
| 5<br>Owner      | ( 0.14                            | 7.17 )                    | ( 0.13                            | 7.95 )                    |
| 9<br>Owner      | 0.13 ↓                            | 7.50 ↑                    | 0.12 ↓                            | 8.3 ↑                     |
| For 500<br>Node | 5 SCN                             |                           | 10 SCN                            |                           |
|                 | Average<br>Transaction<br>Latency | Transaction<br>Throughput | Average<br>Transaction<br>Latency | Transaction<br>Throughput |
| 1<br>Owner      | ( 0.375                           | 2.66 )                    | ( 0.39                            | 2.53 )                    |
| 23<br>Owner     | 1.57                              | 0.63                      | 1.37                              | 0.73                      |
| 45<br>Owner     | 1.73 ↑                            | 0.57 ↓                    | 1.9 ↑                             | 0.51 ↓                    |

Results are expressed in seconds.

Fig. 6. AvgTL and TT results for the data request transaction.

Typically, outcomes for 100 nodes should be similar to results for 500 nodes. This is because the workload is not high for 100 nodes. Also, for 100 nodes, the system occupancy rate is low. Consequently, with the simultaneous escalation of both the system workload and occupancy rate at a specific pace, the outcomes of the system exhibit increased consistency. Another noteworthy aspect is that, despite the identical transaction numbers for the components marked with blue and red parentheses in Figure 6, their results are disparate. In essence, initiating five data sharing requests through a single SCN and submitting a singular data sharing request through five distinct SCNs yield distinct outcomes.

It is discerned that, in scenarios where the total number of transactions remains constant, an increase in the number of SCNs making the request leads to a rise in AvgTL and a reduction in TT values.

Figure 7 depicts the AvgTL and TT results for data request transactions with collisions based on particular workloads. Four workloads are applied to 100 nodes and 500 nodes, as described below:

| For 100                           | 10% Collision |       | 25% Collision |       | 50% Collision |       | 100% Collision |       |
|-----------------------------------|---------------|-------|---------------|-------|---------------|-------|----------------|-------|
| Node                              | ATL           | TT    | ATL           | TT    | ATL           | TT    | ATL            | TT    |
| 2 SCN                             |               |       |               |       |               |       |                |       |
| 5 Owner                           | 0.61          | 1.645 | 0.64          | 1.56  | 0.66          | 1.51  | 0.63           | 1.58  |
| 9 Owner                           | 0.51          | 1.96  | 0.53          | 1.89  | 0.54          | 1.85  | 0.53           | 1.92  |
| 10 SCN                            |               |       |               |       |               |       |                |       |
| 23 Owner                          | 1.96          | 0.51  | 1.98          | 0.50  | 2             | 0.49  | 2.05           | 0.48  |
| 45 Owner                          | 2.43          | 0.41  | 2.573         | 0.388 | 2.75          | 0.362 | 2.86           | 0.349 |
| Results are expressed in seconds. |               |       |               |       |               |       |                |       |
| ATL: Average Transaction Latency  |               |       |               |       |               |       |                |       |
| TT: Transaction Throughput        |               |       |               |       |               |       |                |       |

Fig. 7. AvgTL and TT results for the data request transaction with collisions.

- (1) At the same time, 20% of the SCNs in the system, respectively, buy the genomic data of 5% and 10% of the data owners with 10% collision.
- (2) At the same time, 20% of the SCNs in the system, respectively, buy the genomic data of 5% and 10% of the data owners with 25% collision.
- (3) At the same time, 20% of the SCNs in the system, respectively, buy the genomic data of 5% and 10% of the data owners with 50% collision.
- (4) At the same time, 20% of the SCNs in the system, respectively, buy the genomic data of 5% and 10% of the data owners with 100% collision.

With an increase in the volume of transactions, the results for 100 nodes demonstrate a concurrent decrease in AvgTL and a rise in TT.

Conversely, an analysis of 500 nodes reveals that as the number of transactions expands, AvgTL experiences an increase while TT undergoes a decline. This pattern is consistent with the depiction in Figure 6. It is noteworthy that the workload for 100 nodes is characterized by a degree of moderation, and the network occupancy rate is at a moderate level. Therefore, if both the network workload and occupancy rate increase simultaneously, the system is expected to yield more reliable results. The primary objective was to determine agunetwork performance if many SCNs simultaneously purchased the same data. As can be observed, as the frequency of collisions in both workloads increases, the AvgTL rises, and TT decreases.



|                                   | For 100 Node | 5 Owner | 9 Owner | 45 Owner | 90 Owner |
|-----------------------------------|--------------|---------|---------|----------|----------|
| 2 SCN                             |              |         |         |          |          |
| 100 %Collision                    |              |         |         |          |          |
| Average Transaction Latency       | 0.63         | 0.53    | 1.96    | 2.18     |          |
| Transaction Throughput            | 1.58         | 1.92    | 0.51    | 0.46     |          |
| Results are expressed in seconds. |              |         |         |          |          |

Fig. 8. AvgTL and TT results for the data request transaction, with 100% collisions.

Finally, in Figure 8, simultaneously, 20% of the SCNs in 100 nodes, respectively buy the genomic data from 5%, 10%, 50%, and 100% of the data owners with 100% collision. This assessment aims to evaluate system performance in response to an escalating number of requested data and increasing overlap. Analysis of the outcomes for genomic data at 5% and 10% reveals a trend where, as the transaction volume grows, there is a decrease in AvgTL coupled with an increase in TT. Conversely, when scrutinizing results for genomic data at 50% and 100%, it is evident that with a rise in the number of transactions, AvgTL increases while TT decreases. For the genomic data at 5% and 10%, characterized by a low workload and a correspondingly low system occupancy rate, an increase in both the system workload and occupancy rate at a certain pace is anticipated to yield more consistent results. In a broader context, the general observation is that with a 100% collision scenario and an upsurge in transaction volume, AvgTL tends to increase while TT decreases.

If GenShare can provide secure aggregation and basic statistical operations over genomic data with the use of a PHE (e.g., Paillier) [55], this is under the constraint of needing an additive, but not arbitrary mathematical operation—For example: statistics such as mean and standard deviation of a dataset are supported by Paillier, while gene–gene interactions (epistasis) or polygenic trait estimation and dynamic re-annotation based on non-coding regulatory elements are not. Furthermore, due to the fact that the genomic data is hashed and encrypted upon submission, there is currently no out-of-the-box way to securely update the dataset with new genetic markers being discovered. One possible way around would be for data owners to reprocess and reload their whole dataset with the new markers. This leaves data privacy intact without need of architectural changes, but burdens the user computationally and in terms of ease of use—particularly for large datasets or users who are needing to discover markers often. Therefore, a potential higher-scale solution would be to incorporate FHE in the future to be able to analyze behavior in a more expressive manner or develop privacy-preserving data versioning to safely propagate marker updates. We view this as an open problem and a potential direction for future work.

## 7 Conclusion

In this article, the GenShare platform is proposed, which is a hybrid framework based on Hyperledger Fabric, Ethereum-derived GenShare, HE, and Intel SGX technologies. The objective of the platform is to enable data owners and researchers to share genomic data in a manner that protects their privacy and to do computations on genomic data using SCNs. GenShare protects the privacy and security of genomic data by combining HE with SGX and BC. Furthermore, it provides BC-driven resolutions addressing issues such as analysis expenses, data ownership, data acquisition methodologies, and a diverse array of other challenges associated with data. The hybrid platform GenShare encompasses researchers, data proprietors, and secure computing nodes. GenShare enables data owners to privately share their data indexes on the BC [27], whereas researchers are a public

community that wishes to access shared data. Thus, GenShare has selected both Hyperledger Fabric and Ethereum to fulfill these needs.

For performance analysis of the system, the Hyperledger Fabric network is constructed. The system's performance has been demonstrated at a variety of workload rates. The following results will be collected in time as all such findings are analyzed. Firstly, due to the impact of internet speed on the results of the worldwide network test, a continuous rise rate should not be anticipated. As transaction volume increases, AvgTL increases and TT falls, although transaction volume is not the only factor that influences metrics. In instances where the message size and the volume of nodes initiating requests for the procurement of genomic data experience augmentation, the AvgTL exhibits an increase, whereas the TT demonstrates a decrease for two distinct workloads characterized by an identical aggregate number of processes. Significantly, with a simultaneous increase in both the workload and occupancy rate at a specific rate, the system achieves an augmented capability to yield more dependable results.

In the future, additional enhancing of system robustness and trustworthiness are expected. Future research will concentrate on how to make the platform address trust even more explicitly. Especially, we expect that the combination of zero-knowledge proof-based mechanisms with verifiable computation techniques could provide computation verification while still keeping secret. Auditor nodes will be treated as standalone agents to log and verify the behavior of SCNs for improved transparency and accountability. Moreover, the use of advanced threshold cryptography will assist collaborative validation and access control. These strategic improvements will further enhance platform's trust model and result in a more robust secure environment to share and analyze genomic data. Further work can also investigate the combination of FHE and privacy-preserving update techniques to support richer and dynamic genomic analysis, particularly when considering novel genetic markers and combinatorial traits.

These technical enhancements are designed not only to strengthen computing power but also to bolster the ethical underpinning of the platform. We recognize the ethical dilemma that if not communication transparently, the partial-featured systems might have the potential for misuse or being misrepresentations in the cases where actors are dealing with private genomics data. GenShare was not designed to meet all the requirements of genomic research, but rather to serve as a proof-of-concept that secure, privacy-preserving data sharing, and computation can be facilitated for a subset of clinically and scientifically relevant tasks. We have specifically detailed these limitations and highlight that the transparency of system capabilities is a fundamental design philosophy. In future iterations, we hope to investigate cryptographic protocols that are able to do more, for example, FHE and secure update mechanisms, in the interest of growing GenShare into a more complete, ethically sound tool.

Lastly, besides the crypto and architectural improvements, we also pursue the increase of the platform's real-world relevance. Finishing the missing features and making the middleware more robust, we will be able to test the efficiency and scalability of GenShare in more realistic settings.

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